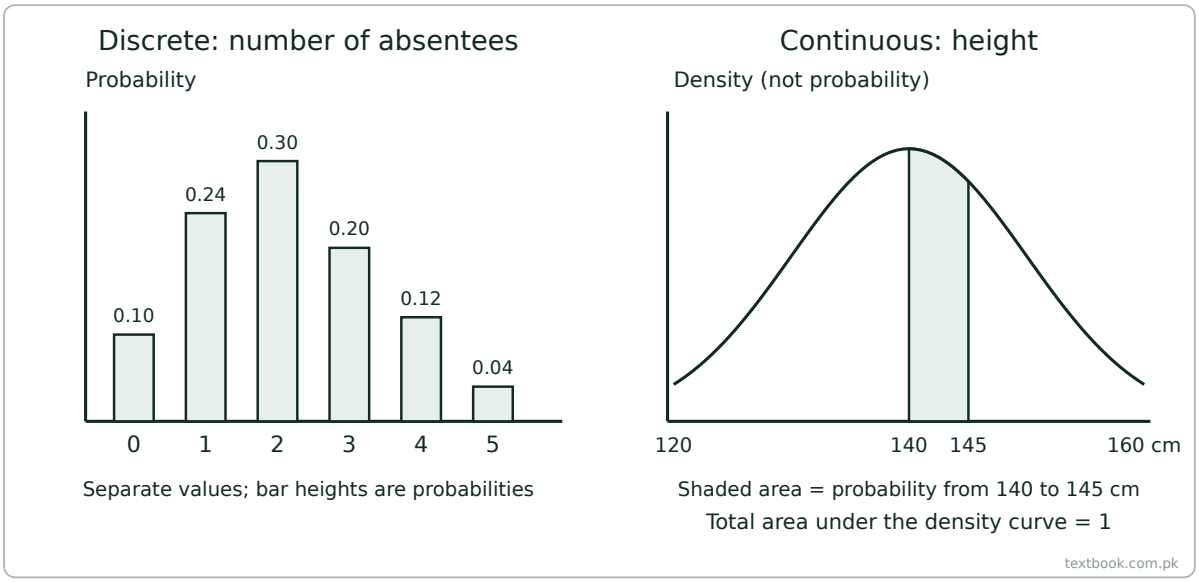
Draft - expert review pending

Note: This unit is complete and passes the platform's automated checks, but it is not reviewed yet. So, treat its sources and claims with care. You are encouraged to report errors and provide feedback.

Topic 10.2 - Probability distributions

A real classroom situation

A head teacher in Larkana must decide how many substitute worksheets to print for a class of 40 each day, one for every absent pupil. The class teacher has counted the absentees on the last 50 school days: no one was absent on 5 days, one pupil on 12 days, two on 15 days, three on 10 days, four on 6 days and five on 2 days. The head teacher asks two questions. "How likely is each number of absentees?" and "On average, how many worksheets will we need per day?" A probability distribution answers the first; its expected value answers the second.



Explanation

What a probability distribution is

The **probability distribution** of a discrete random variable lists every value the variable can take together with the probability of each value (Illowsky & Dean, 2023, Section 4.1). It can be written as a table, drawn as a bar chart with one bar per value, or given as a formula. We write $P(X = x)$ for "the probability that X takes the value x".

> **Worked example: equally likely outcomes.** Toss two coins and let X be the number of heads. From Topic 10.1, the four equally likely outcomes give $X = 2$ once, $X = 1$ twice and $X = 0$ once. So:

x	0	1	2
$P(X = x)$	1/4 = 0.25	2/4 = 0.50	1/4 = 0.25

Building a distribution from records

All school figures in this unit are invented teaching data, not findings about a named school's attendance. If we choose one of the 50 recorded days at random, relative frequencies give its exact distribution. Using that table for a future day is an estimate: we assume conditions remain similar. A holiday, illness outbreak or change in class size may make it unsuitable. The table gives zero observed frequency for 6 to 40 absentees, not proof that these counts are impossible on future days.

When outcomes are not equally likely, we use records, as Unit 9 did with attendance registers. Divide each frequency by the total.

> **Worked example: absentees in Larkana.** Let X be the number of

> pupils absent on a school day. Over 50 days:

x **(absentees)**	**0**	**1**	**2**	**3**	**4**	**5**	**Total**
Days	5	12	15	10	6	2	50
$P(X = x)$	5/50 = 0.10	0.24	0.30	0.20	0.12	0.04	1.00

The distribution shows at a glance that 2 absentees is the most likely number, and that 4 or more absentees happen on 0.12 + 0.04 = 0.16 of days.

> **Common misconception.** Many trainees think "a random variable is always evenly distributed", so that every value is equally likely. That is true for the score on one fair die, but not for absentees: 2 absentees (0.30) is seven and a half times as likely as 5 absentees (0.04). Even two fair coins give 1 head twice as often as 0 heads. Each random variable has its own distribution, and finding its shape is the point of the exercise.

Two checks every valid distribution passes

A table is a valid probability distribution only if it passes both checks (Illowsky & Dean, 2023, Section 4.1, NIST/SEMATECH, Section 1.3.6.1):

1. **Each probability lies between 0 and 1**, inclusive. No probability can be negative or greater than 1.
2. **The probabilities add up to exactly 1.** The list covers every possible value, so one of them must happen.

> **Worked example.** A trainee proposes this distribution for Y, the number of correct answers out of 3:

y	0	1	2	3
$P(Y = y)$	0.1	0.3	0.4	0.3

Check 1: every value lies between 0 and 1, so it passes. Check 2: 0.1 + 0.3 + 0.4 + 0.3 = 1.1, so it fails. The table is **not** a valid distribution; something has been counted twice or a probability is too large. Always run *both* checks: a table with probabilities 0.5, 0.6 and -0.1 adds to 1 but fails check 1.

x (absentees)	0	1	2	3	4	5	Total
Days	5	12	15	10	6	2	50
P(X = x)	0.10	0.24	0.30	0.20	0.12	0.04	1.00
x × P(X = x)	0	0.24	0.60	0.60	0.48	0.20	2.12

✓ Check 1: every probability is between 0 and 1

✓ Check 2: probabilities add to 1

Expected value E(X) = 2.12 absentees per day

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The expected value

The **expected value** of a discrete random variable, written $E(X)$, is the mean of its distribution. Multiply each value by its probability and add the results (Illowsky & Dean, 2023, Section 4.2):

$E(X)$ = sum of $x \times P(X = x)$

> **Worked example.** For the Larkana absentees:

$E(X) = 0 \times 0.10 + 1 \times 0.24 + 2 \times 0.30 + 3 \times 0.20 + 4 \times 0.12 + 5 \times 0.04 = 0 + 0.24 + 0.60 + 0.60 + 0.48 + 0.20 =$ **2.12**

This is a weighted mean, like the weighted mean of Unit 4: each value is weighted by how often it happens. It means that over many days the school should expect *about* 2.12 absentees per day on average, so about 212 worksheets for every 100 days.

> **Common misconception.** Many trainees think "the expected value is the value you will most likely see". It is not. No single day can have 2.12 absentees, and the most likely value is 2 (probability 0.30), which is the mode of the distribution. The expected value is a *long-run average* over many repetitions (Illowsky & Dean, 2023, Section 4.2). For two coins the expected number of heads is $0 \times 0.25 + 1 \times 0.50 + 2 \times 0.25 = 1$, which here happens to be the most likely value too, because this particular distribution is symmetric with its highest probability at the centre. Symmetry alone does not guarantee that a mean is the most likely value.

Distributions of continuous random variables

A continuous random variable, such as a pupil's height, cannot be listed value by value. Its distribution can be represented by a density curve over the possible values; the curve need not be smooth or bell-shaped. The probability of a value falling in an interval is the *area* under the curve over that interval, and the whole area under the curve equals 1, the continuous form of check 2 (NIST/SEMATECH, Section 1.3.6.1). The probability of one exact value, such as exactly 140.000 cm, is zero; only intervals carry probability. The height of this curve is a density, not a probability. It may exceed 1; areas, which are probabilities, cannot exceed 1.

Activity: Two-coin experiment

Work in groups of three. About 20 minutes.

1. One member tosses two coins 20 times. Another records the number of heads (0, 1 or 2) each time. The third checks the tally.
2. Turn your tally into a distribution: divide each count by 20. Run both checks on your table.
3. Compute the expected number of heads from your own table.
4. Compare your table and expected value with the theoretical ones (0.25, 0.50, 0.25, expected value 1). Pool the results of all groups in the class and compare again. Discuss whether the pooled table is closer. A larger set often gives a closer estimate, but it is not guaranteed on any one trial.

Check your understanding

1. **Remember** State the two conditions a table must meet to be a valid probability distribution.

2. **Understand** Explain why the expected number of absentees can be 2.12 even though 2.12 pupils can never be absent.

3. **Apply** A spinner gives 1 point with probability 0.5, 2 points with probability 0.3 and 5 points with probability 0.2. Compute the expected number of points.

4. **Apply** Over 20 days a canteen sold 0 extra samosas on 4 days, 1 on 10 days and 2 on 6 days. Build the probability distribution and check it.

Summary

A probability distribution lists every value of a random variable with its probability. It can come from equally likely outcomes or from records. A table is valid only if every probability lies between 0 and 1 and the probabilities add up to 1. The expected value multiplies each value by its probability and adds: it is the long-run average, not the most likely value. A continuous random variable can be represented by a density curve, and probability is area under it.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can build the probability distribution of a discrete random variable from a frequency table.
- ☐ I can check whether a table is a valid probability distribution using both conditions.
- ☐ I can compute an expected value and explain why it is not the same as the most likely value.

Try this at your practicum school

With your cooperating teacher's permission, count the number of pupils absent in one class on each of the last 20 school days, using the register. Build the probability distribution, run both checks, and compute the expected number of absentees. Share one sentence with the teacher about what the expected value means for planning.

Summative task

A school librarian in Nawabshah gives two tables for X, the number of books a pupil borrows in a week.

x	0	1	2	3
Table A: $P(X = x)$	0.35	0.40	0.15	0.10
Table B: $P(X = x)$	0.20	0.50	0.40	-0.10

a) Run both checks on each table and decide which is a valid probability distribution. b) For the valid table, compute the expected value. c) In 80-120 words, analyze what the librarian can and cannot conclude from the expected value when ordering new books for 300 pupils, and explain why checking only that the probabilities add to 1 would have missed the problem in the other table.

Mini-rubric

Criterion	Levels
Checks applied (Apply)	0-3: both checks run on both tables with the right verdicts / one table right / checks named but misapplied.
Expected value (Apply)	0-2: correct value with working / method right, arithmetic slip / incorrect.
Interpretation and validity (Analyze)	0-3: treats the expected value as a long-run average for planning, notes it is not a value any pupil borrows, and explains why one check alone is not enough / partly / not at all.

Further reading

Illowsky, B., & Dean, S. (2023). *Introductory Statistics 2e.* OpenStax. https://openstax.org/books/introductory-statistics-2e/pages/4-1-probability-distribution-function-pdf-for-a-discrete-random-variable



openstax.org/books/introductory-statistics-2e/pages/4-1-probability...

- Section 4.1, the two characteristics of a discrete probability distribution.

Illowsky, B., & Dean, S. (2023). *Introductory Statistics 2e.* OpenStax. https://openstax.org/books/introductory-statistics-2e/pages/4-2-mean-or-expected-value-and-standard-deviation



openstax.org/books/introductory-statistics-2e/pages/4-2-mean-or-exp...

- Section 4.2, the expected value as a long-run average.

NIST/SEMATECH. *e-Handbook of Statistical Methods*, Section 1.3.6.1, "What is a Probability Distribution". https://www.itl.nist.gov/div898/handbook/eda/section3/eda361.htm



itl.nist.gov/div898/handbook/eda/section3/eda361.htm

Lane, D. M., & Ziemer, H. (n.d.). *Distributions.* Online Statistics Education. Rice University. https://onlinestatbook.com/2/introduction/distributions.html



onlinestatbook.com/2/introduction/distributions.html

- "Distributions", including the continuous case.

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